Catch the exit before it happens

A profit-targeted churn-retention engine for Company A (US wireless)

Proof-of-Concept proposal for the executive team

GCI World 2026 Winter Final Assignment | Matsuo-Iwasawa Lab, The University of Tokyo

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Executive summary: churn is predictable at two lifecycle moments, and targeting by profit makes it pay

The problem

A mature US market (1.7 connections per person) means growth must come from **keeping** customers. Carriers lose about 1% (postpaid) to 3% (prepaid) of customers every month.

What we found

Churn jumps from **33% to 58%** at months 10-12 and from **39% to 58%** once the handset is 300-420 days old.

What we propose

A scoring engine that picks **who to contact and when**, only where the expected profit is positive, starting with an 8-week randomised pilot.

0.698

ROC-AUC of the LightGBM model on 20,000 unseen customers (baseline 0.678)

$49,526 / month

net profit per 100,000 customers at base-case assumptions (about $594,309 a year)

16,639 contacts

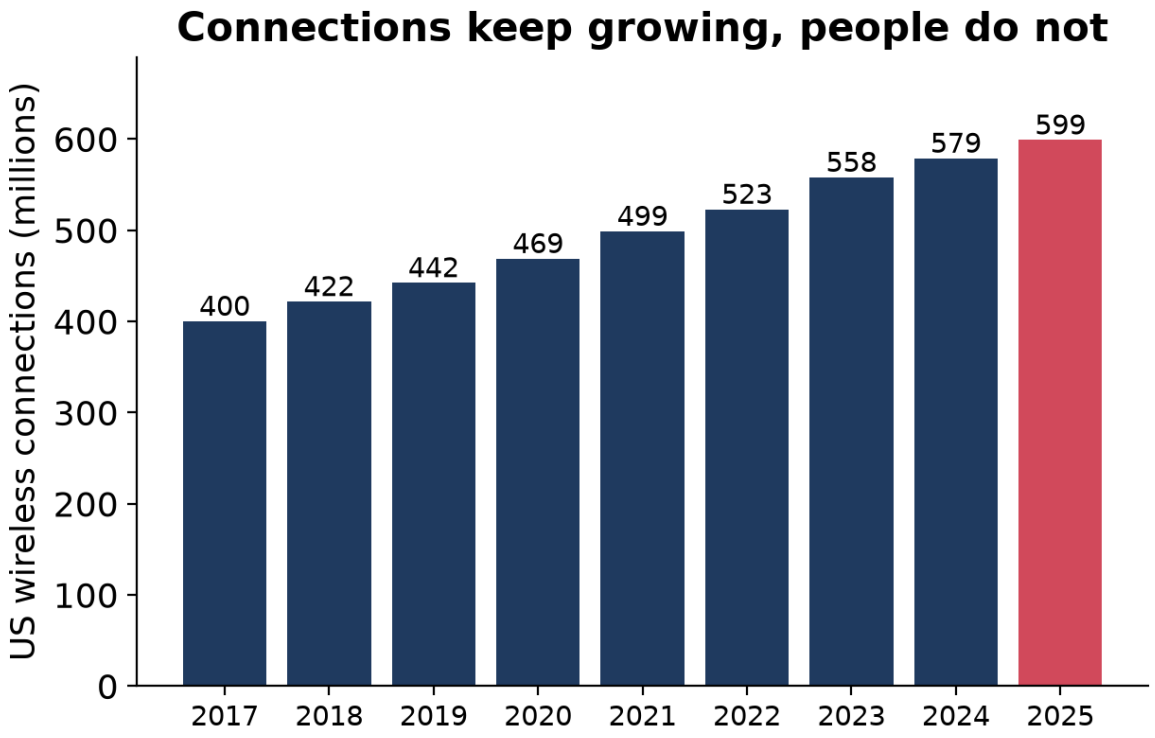
out of 100,000 customers get an offer; contacting everyone would lose $153,622 a month

Why it matters

The model alone is not the product, **offer design is**. Profit falls from $122,918 a month when few loyal customers claim the offer (5%) to $9,686 when all of them do, and the pessimistic scenario is only break-even ($90). The pilot therefore measures offer take-up before any scale-up.

Sources: [3][4][5] market churn; own analysis of Company A data; [8] cost-benefit framework

Market: a saturated US market makes keeping a customer worth more than winning one



Note: 2025 total from Light Reading citing CTIA; 46% of 2025 connections are not phones [2].

0.9-1.0%
monthly postpaid phone churn at T-Mobile, Verizon, AT&T (2025) [3][4][5]

2.7-3.7%
monthly prepaid churn at the same carriers, about 3x higher [3][4][5]

5-25x
cost of acquiring vs retaining a customer across industries (HBR) [6]

~3x
$160 acquisition vs $53 retention per quarter in 41 countries' wireless data [7]

So what for Company A

- Connections grew ~5% a year but already exceed the population, so net adds now come from devices and rivals' customers [1][2].
- Churn rises as customers roll off device-financing plans and carriers answer with large discounts [12], so timing matters.

Sources: [1] CTIA 2025; [2] Light Reading 2026; [3][4][5] company reports; [6] HBR 2014; [7] J. Marketing Research

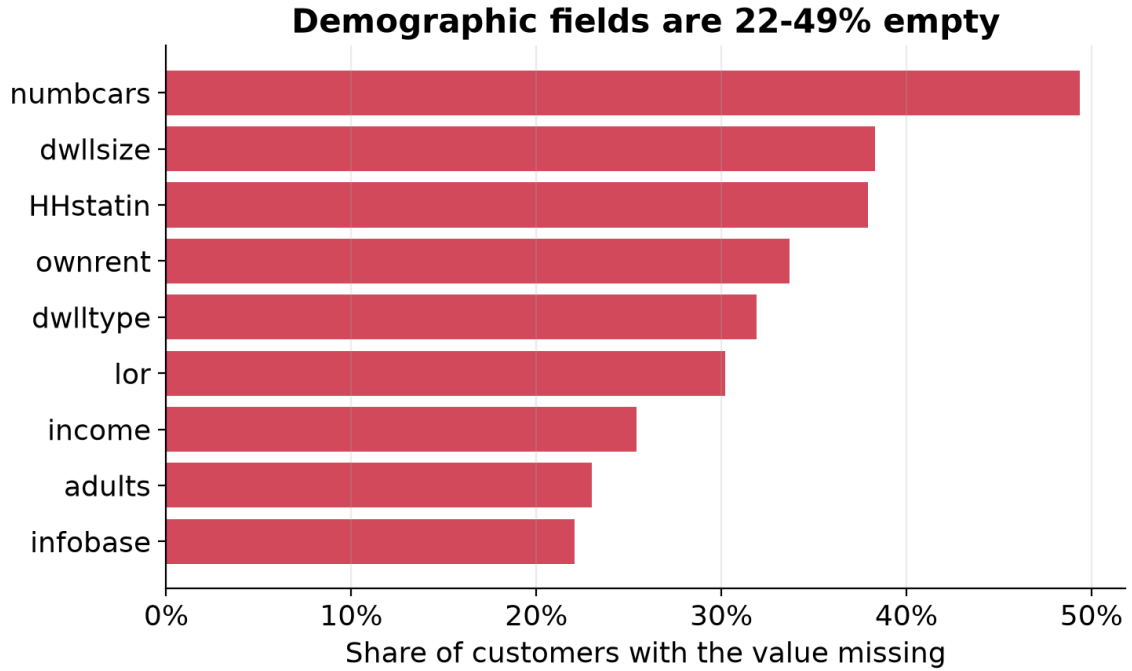
The client and the data: 100,000 customers, 98 fields, one clear target

Situation

- Company A is a US wireless carrier with a large usage and billing history but **no in-house ML capability** [16].
- Two linked tables (Client, Record), one row per customer, joined on Customer_ID (1:1, no duplicates).
- Contents: tenure, plan price, usage and call quality, handset history, credit class, region and household demographics.

Why churn as the target

- It is the direct driver of revenue loss and acquisition spend, it is observed for every customer, and the market data above shows retention is the cheapest lever [6][7].
- Churn here = left within 31-60 days of the observation date.

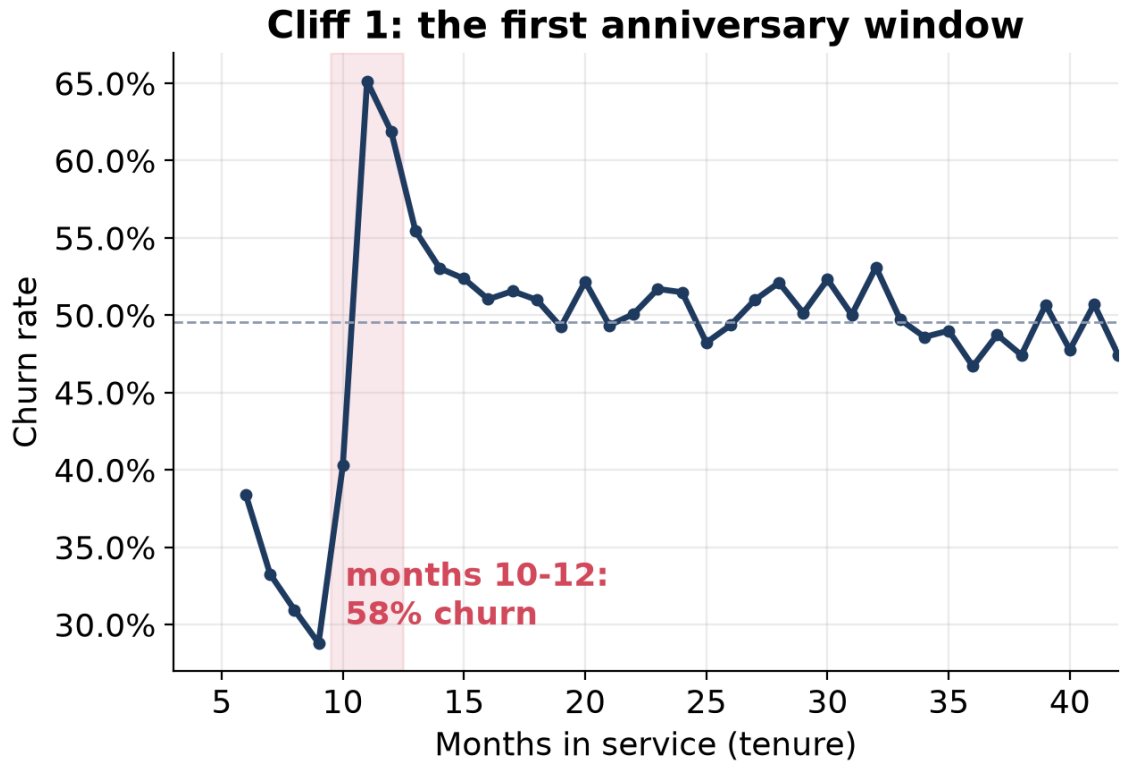


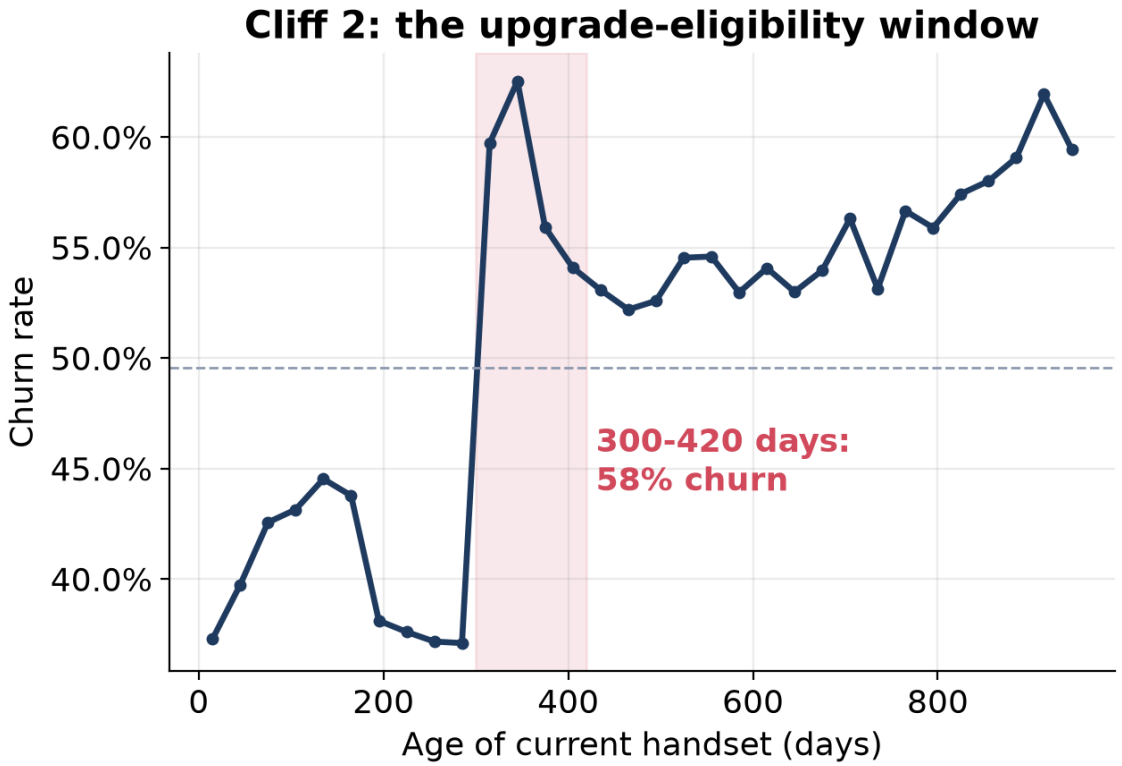
Two data cautions

- **Over-sampled target:** 49.6% of the sample churned, but real carriers lose ~1-3% a month. Classic practice for model building [9]; we re-weight to a realistic rate before computing money.
- **Sparse demographics:** 22-49% missing; we keep the signal by counting how many fields are missing.

Sources: Company A dataset overview and tutorial [16]; [9] on over-sampled churn samples; own analysis

EDA 1: churn jumps from 33% to 58% at month 10, and again when the handset is ~300 days old





What it means

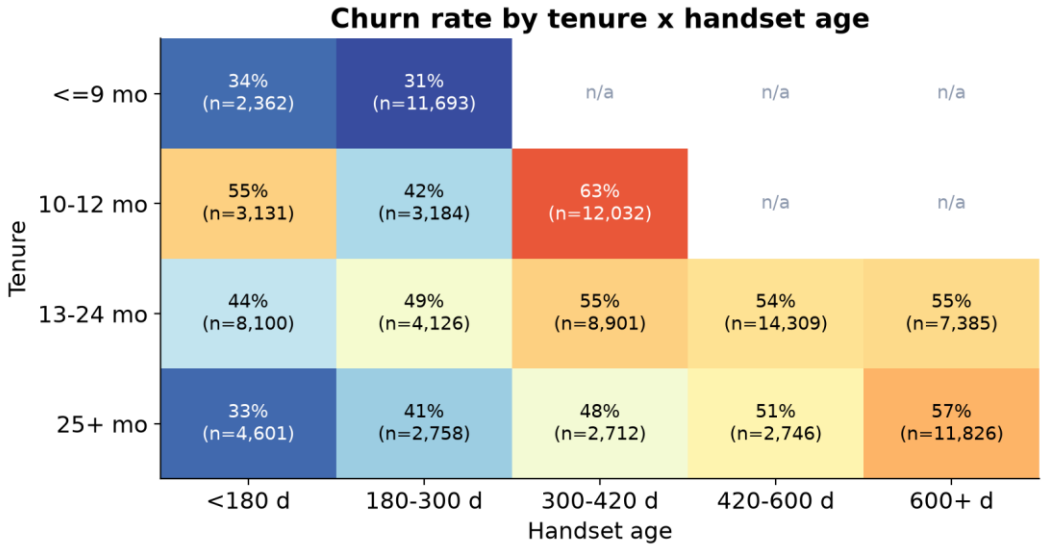
- Churn is not spread evenly: it clusters in **two predictable windows** in the customer's lifecycle - the first anniversary and the handset-upgrade window.

Hypothesis (to be tested in the pilot)

- Customers re-evaluate their carrier when a contract term or financing period ends, or an upgrade becomes available [11][12]. In a Canadian study, customers who upgraded their handset once were 22.5% less likely to churn [11].

Sources: Own analysis of Company A data (n = 100,000); [11][12] on contract/upgrade timing

EDA 2: the two windows compound, while service quality and care calls explain little



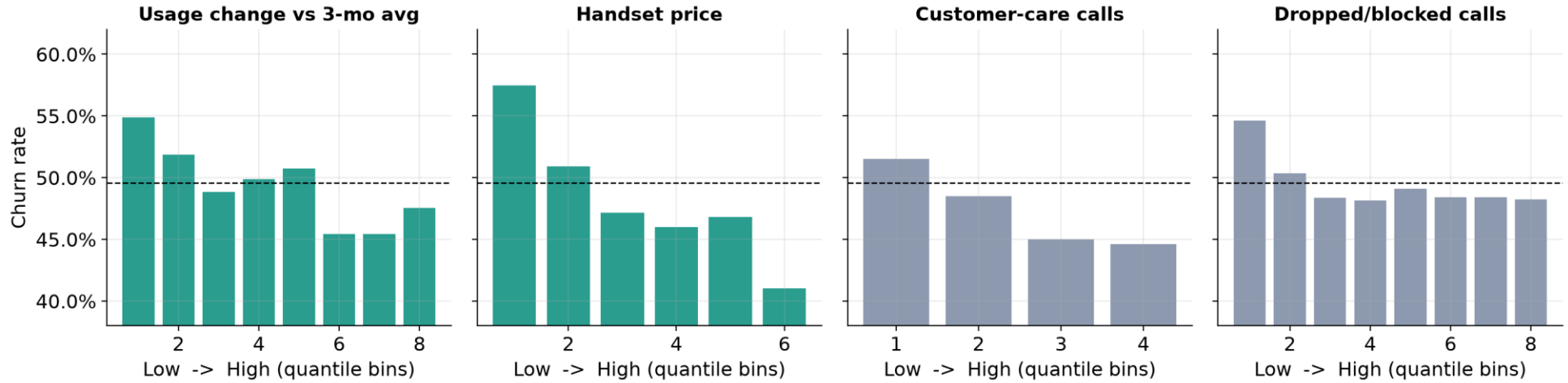
Reading the lower charts

- Falling usage and cheaper handsets go with churn (gap between the lowest and highest group: 9 and 16 percentage points).
- Care calls and dropped/blocked calls move it less (7 and 6 points): network quality is a secondary story here.

Takeaway

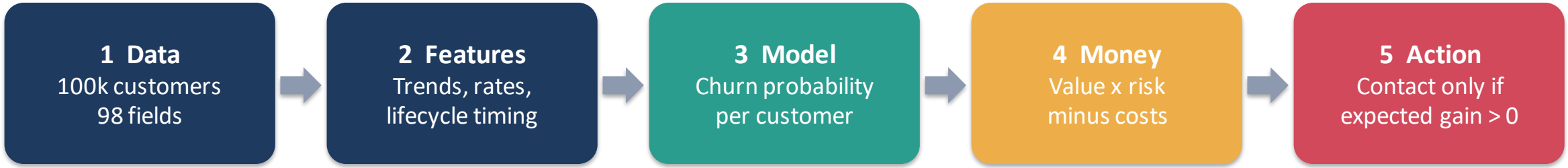
- Customers 10-12 months in with a 300-420 day-old handset churn at 63% (12,032 customers). The same tenure band with a newer handset churns much less.
- Regions differ too (Midwest Area 46% vs Northwest/Rocky Mountain Area 57%).

Business implication: spend on timing and offers, not on network fixes, for this problem.



Sources: Own analysis of Company A data (n = 100,000)

Problem definition: rank customers by churn risk, then by expected profit



Business objective

- Maximise the net profit of a monthly retention campaign. The contact list is chosen by expected gain.

ML task

- Supervised binary classification: P(churn within 31-60 days).

Target and why

- Churn is the direct driver of revenue loss, it is observed for every customer, and keeping a customer is cheaper than replacing one [6][7].

How we will judge success

- **Model metric: ROC-AUC** - how well it ranks churners above stayers; not distorted by the over-sampled 50/50 sample. Supported by top-decile lift and PR-AUC.
- **Business metric: net profit per 100,000 customers** after re-weighting to a realistic churn rate (2% base case).
- **Fair testing:** all choices are tuned on 80% of the data with 5-fold cross-validation; a 20% hold-out of 20,000 customers is opened once at the end.

Sources: [8] Verbraken et al. 2013 (cost-benefit framework); own design

Method: simple ratios that describe behaviour, and four model families compared

Feature engineering (98 raw columns to 113 features)

- **Trends:** last 3/6-month usage, calls and revenue vs life-time average (is the customer fading?).
- **Quality rates:** failed and dropped calls per attempt; care calls per call.
- **Lifecycle:** handset age relative to tenure; handsets per year; flags for the two windows.
- **Money:** overage share of bill, revenue per minute, handset price vs plan price.
- **Missing data:** a count of empty demographic fields, so the gaps themselves carry signal.
- **Result: a small gain (LightGBM AUC 0.697 on raw columns, 0.698 with engineered features).**

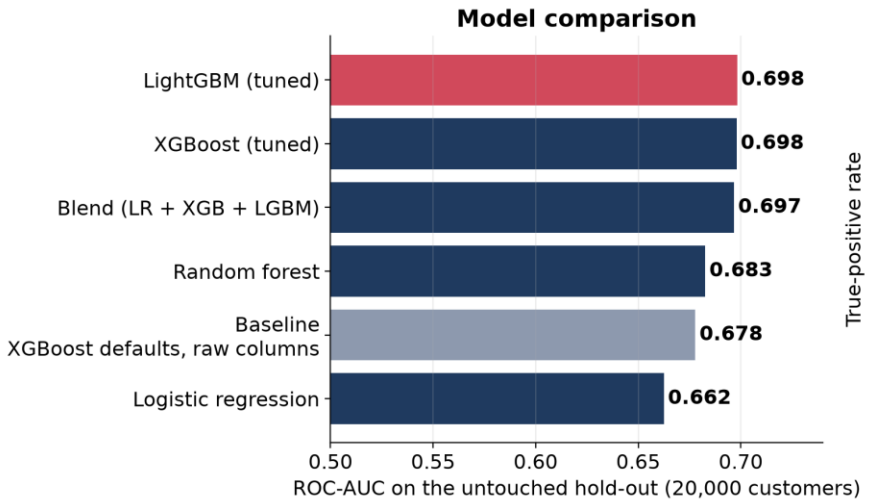
Model	Why we tried it
Logistic regression	Transparent linear baseline the client can audit
Random forest	Many decision trees voting; robust default
XGBoost	Boosted trees; learns interactions such as tenure x handset age
LightGBM	Fast boosted trees with native categories
Blend	Average of the three best, to reduce variance

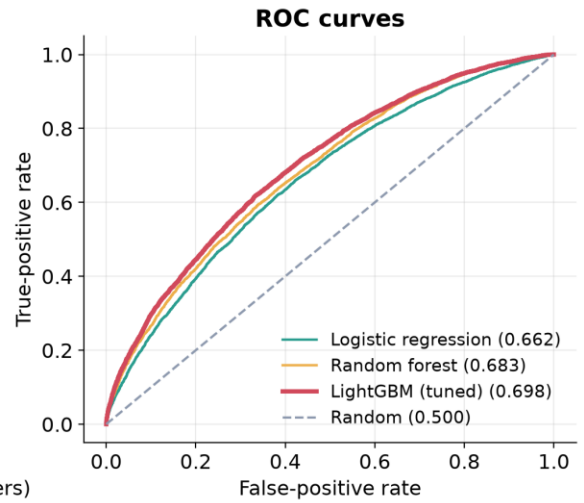
Training protocol

- 80% development / 20% untouched hold-out (stratified, seed 42).
- Hyper-parameters (tree size, number of rounds) chosen by 5-fold cross-validation with early stopping on the development set only.
- Reference = default XGBoost on raw columns, i.e. the tutorial approach.

Sources: [13] LightGBM; [14] XGBoost; [15] scikit-learn; own design

Results: LightGBM and XGBoost tie at a moderate ROC-AUC of 0.698 on unseen customers, 0.020 above the baseline





Model	Hold-out ROC-AUC	5-fold CV AUC	Lift top 10%
Baseline - XGBoost defaults, raw columns	0.678	0.676 ± 0.005	1.56x
Logistic regression	0.662	0.662 ± 0.004	1.49x
Random forest	0.683	0.681 ± 0.005	1.57x
XGBoost (tuned)	0.698	0.698 ± 0.005	1.59x
LightGBM (tuned)	**0.698**	**0.698 ± 0.004**	**1.61x**
Blend (LR + XGB + LGBM)	0.697	0.697	1.60x

How to read it

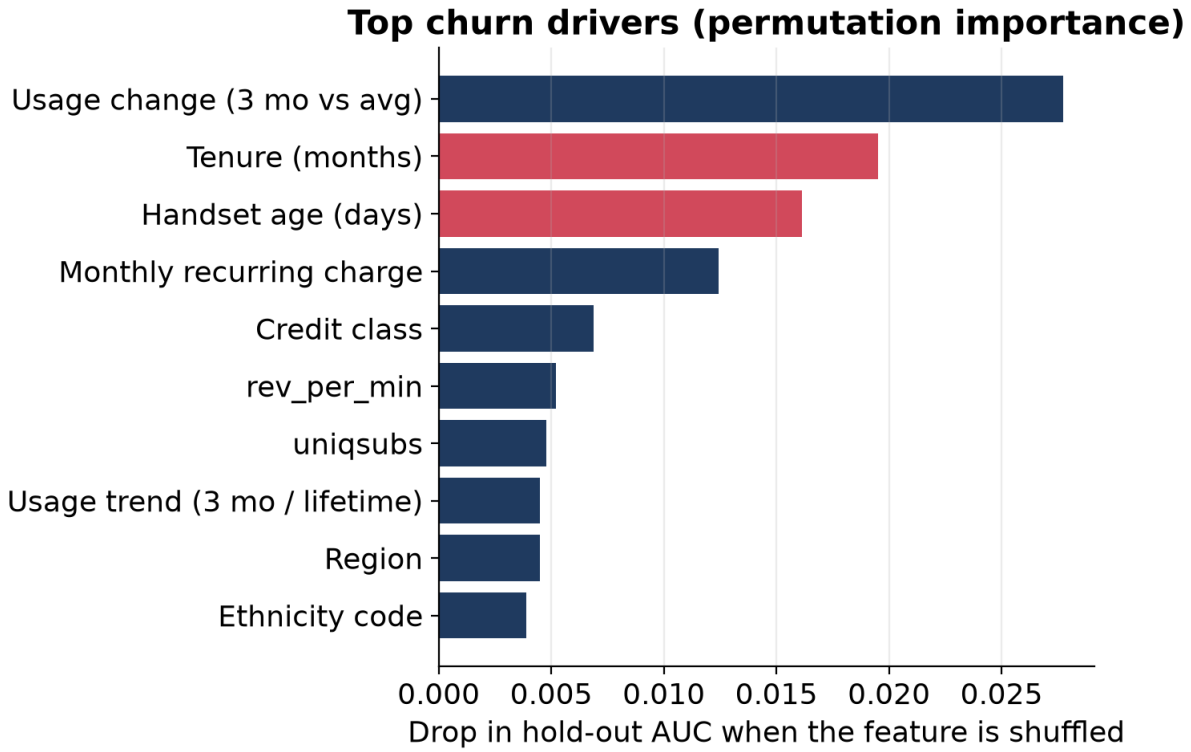
- **Chosen model: LightGBM (tuned)** (best out-of-fold AUC, chosen without looking at the hold-out). Hold-out ROC-AUC **0.698** (95% interval 0.691-0.705); top-10% of risk scores contain **1.6x** more churners than average.
- Accuracy at 0.5 cut-off is 64.1% vs 62.6% for the baseline - shown for the tutorial's metric, but AUC is the metric we trust.

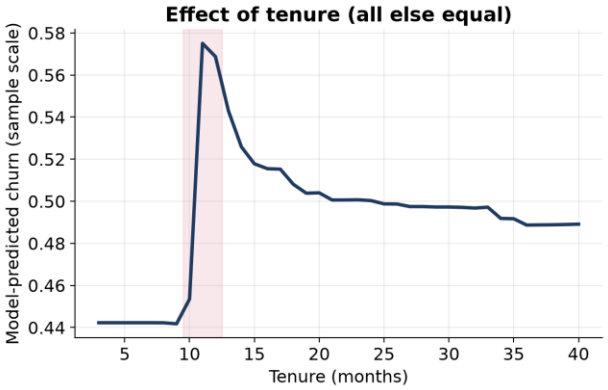
Honest limits

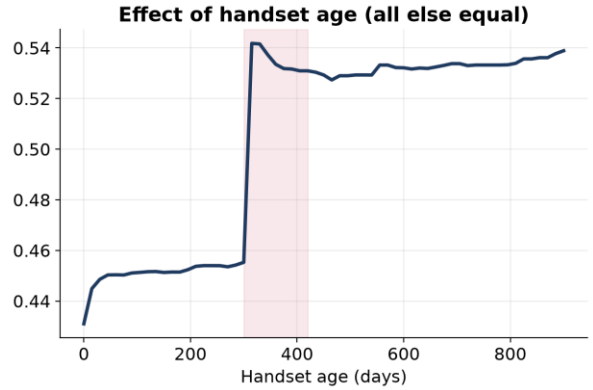
- Behavioural data alone gives a moderate ceiling: published AUCs are ~0.62 on one public carrier dataset [18] and ~0.76 for random forests averaged over eleven telecom datasets [17]; ours is in that range.
- Hence the proposal does **not** rely on a perfect model: it contacts only the few customers where the numbers clearly pay.

Sources: Own analysis; hold-out n = 20,000 customers; [17][18] benchmark context

What drives churn: three of the top drivers are actionable timing levers







From driver to lever

- **Tenure and handset age** (can be acted on): trigger an anniversary check-in ~1 month before month 10, and an upgrade offer before the handset reaches ~300 days.
- **Falling usage** (early warning): score refreshes monthly; a drop in minutes moves a customer up the list.
- **Plan price, credit class, region**: segment and budget rules, not individual actions.
- Not actionable: demographics - used as controls only.

Robustness: the same top drivers appear in gain-based and permutation importance, and in the simple EDA charts - one story, three views.

Sources: Own analysis (permutation importance and partial dependence on hold-out); [11] handset upgrade effect

Business model: contact a customer only if the expected gain beats the cost

Expected gain of contacting customer i =

P(churn) x [P(accepts) x (value kept - offer) - contact cost] - P(stays) x [P(claims anyway) x offer + contact cost]

How the model enters

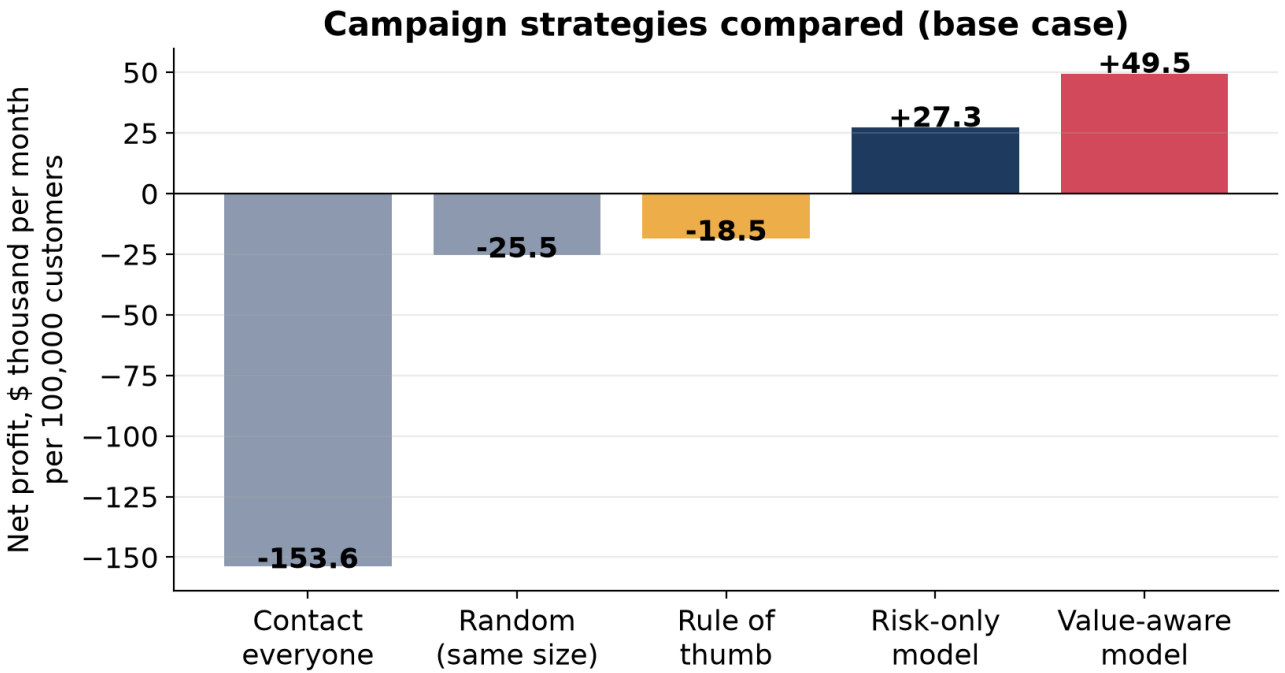
- Churn probability is **calibrated** and re-weighted from the 50% sample to a realistic 2% monthly rate with Bayes' rule.
- Customer value is **individual**: 24 months x 30% margin x the customer's own monthly revenue, so a high-revenue customer justifies a larger list than a low-revenue one.
- Customers are contacted if, and only if, expected gain > 0. No threshold to tune, and it re-optimises when costs change.
- Uncertainty is handled openly: take-up and leakage are the two numbers the pilot must measure.

Assumption	Base	Basis
True monthly churn	2%	Over-sampled sample; carriers report 0.9-1.0% postpaid, 2.7-3.7% prepaid [3][4][5][9]
Churner accepts offer (gamma)	30%	Literature prior mean [8]
Loyal customer claims offer (alpha)	30%	Assumed; literature worst case 100% - see sensitivity
Value of a saved customer	$416	24 months x 30% margin x own revenue; AT&T service EBITDA margin is 57% [5]
Offer / contact cost	$10 / $1	Literature defaults [8][10] (credit; SMS + e-mail)

All of these are inputs, not results: slide 13 shows what happens when they are wrong.

Sources: [8] Verbraken et al. 2013; [10] cost-matrix formulation; [3][4][5] churn; [6][7] cost ratios; own assumptions

Impact: targeting by expected profit nets $49,526 a month per 100,000 customers, while contacting everyone loses $153,622



Per 100,000 customers / month	Value-aware model
Customers contacted	16,639 (16.6%)
Churners reached (of ~2,000)	639
Customers saved (30% accept)	192
Campaign spend	$66,555
Net profit	**$49,526**
ROI on spend	74%
90% interval for profit	$45,779 to $53,931

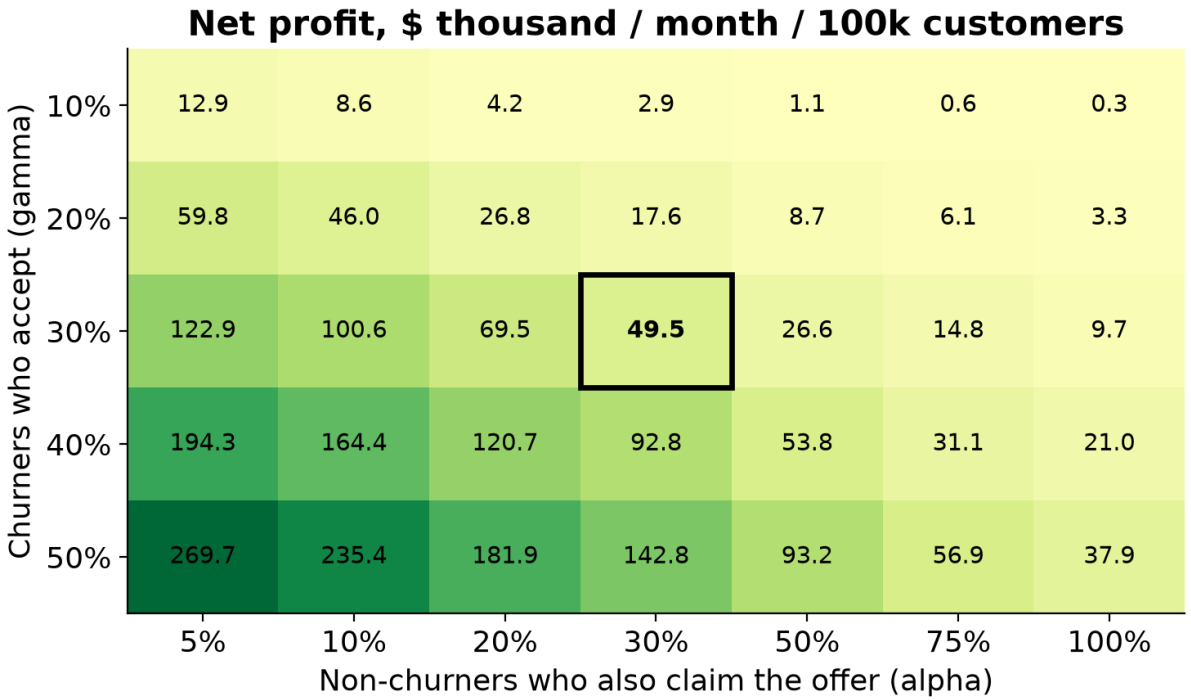
Why not simply contact everyone in the danger windows? The 10-month and 300-day windows raise risk, but at a realistic 2% churn most contacted customers would stay, and every one who claims an offer costs money. The 90% interval reflects hold-out sampling noise only, not uncertainty in the assumptions.

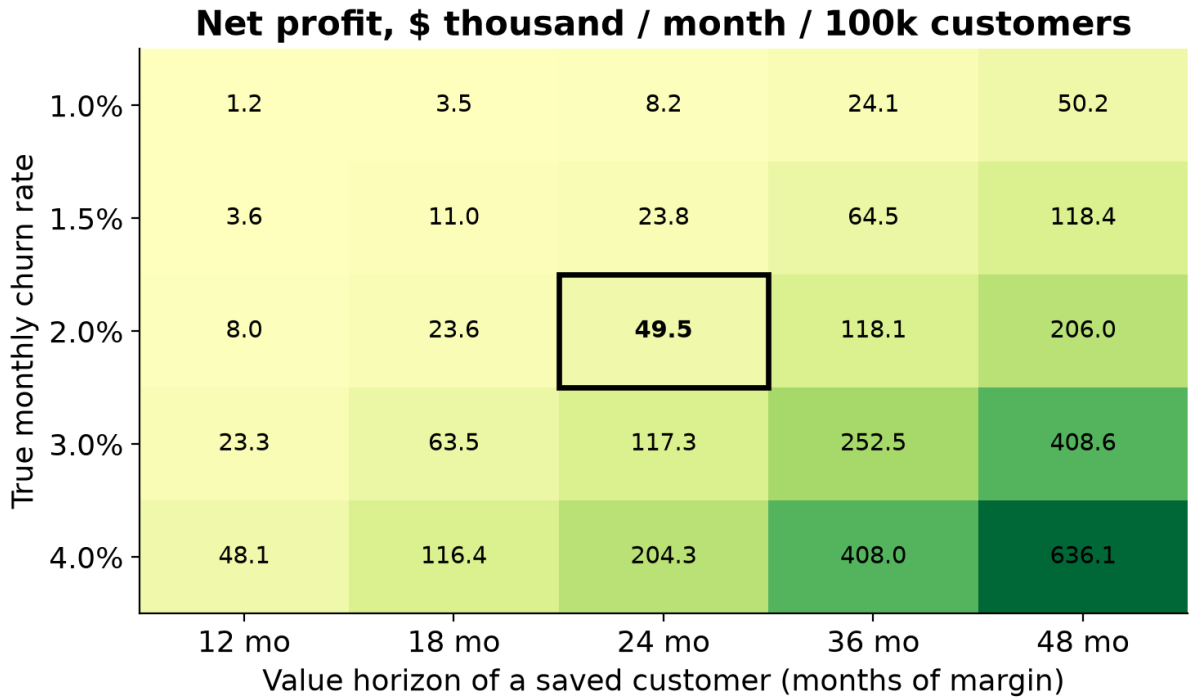
Reading it

- Same offer, same costs: ordering by **risk only** earns $27,256; the **value-aware** list earns $49,526.
- Contacting everyone in a lifecycle window (the rule of thumb, 25,855 contacts) loses $18,506 a month.
- Annualised, base case: about **$594,309** per 100,000 customers (assumes monthly cohorts and a stable save rate).

Sources: Own analysis (hold-out, base case); 90% bootstrap interval; assumptions on slide 11

Sensitivity and risks: offer design decides whether the campaign pays





Scenario	Monthly churn	Churner accepts	Loyal claims offer	Net profit / month
Pessimistic	1.0%	20%	50%	$90
Base	**2.0%**	**30%**	**30%**	**$49,526**
Upside	2.0%	40%	10%	$164,356

Key risks

- Pessimistic case is only break-even: about 22 customers per 100,000 are worth contacting.
- Offer leakage and weak save rates (measure first).
- Correlation is not causation: lifecycle windows may be contract effects; use a randomised hold-out.
- Data age and the 2% rate assumption; recalibrate on live data.

Sources: Own analysis; profits in $ thousand per month per 100,000 customers; boxed cell = base case

Proposal: a lifecycle retention engine, proven in an 8-week pilot before any scale-up

Phase 1: Pilot (weeks 1-8)

- Score the base monthly with the model; build the contact list by expected gain.
- Randomise: 50% of the list gets the offer and 50% is held out, so we can measure the **true save rate** and **leakage**.
- Two plays: an anniversary check-in (month 9) for customers still on their original handset, and an upgrade offer before the handset reaches ~300 days.
- Cost: model run + $1 per contact + $10 per accepted offer.

Phase 2: Scale (months 3-6)

- Replace assumptions with the measured save rate and leakage; re-optimise the list automatically.
- Add channels (app, store, call) with their own costs.
- Refresh the scores monthly.

Phase 3: Optimise (month 6+)

- Uplift models: target customers who respond **because of** the offer, not merely likely churners.
- Test other levers: plan change, equipment financing.
- Quarterly drift review.

KPIs

Saves per 1,000 contacts, offer leakage, net profit per 100k customers, churn in treated vs hold-out

Go / no-go

Scale only if measured profit stays above $0 at the pilot's own take-up numbers (base case $49,526 a month per 100,000 customers)

Budget at risk

At base case about $66,555 a month per 100,000 customers in campaign spend

Sources: Own proposal; [8] profit-driven churn framework; [11][12] on upgrades and renewal windows

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